

LOG6310: Digital Twin - Project Report :
MonDTreal - A Digital Shadow of Montreal's Road
Network

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1 Introduction

Urban transportation networks are complex, large-scale systems that require continuous maintenance, monitoring, and adaptation to ensure efficient and reliable mobility [2, 3]. In major metropolitan areas such as Montreal, road construction and maintenance activities are both frequent and necessary to preserve infrastructure quality, ensure safety, and accommodate long-term urban growth [4]. However, despite their necessity, these activities often introduce substantial disruptions to normal traffic operations. Lane closures, partial blockages, detours, and temporary speed reductions can significantly alter traffic dynamics, leading to congestion, increased travel times, and reduced network efficiency. These disruptions are rarely confined to a single location, and instead, they tend to propagate through the network, influencing traffic conditions on adjacent and even distant road segments [5].

One of the central challenges in urban transportation management is the ability to understand and quantify the system-wide effects of such localized disruptions. Road networks are inherently interconnected, meaning that changes in one segment can trigger cascading effects across multiple routes due to rerouting behaviour, bottleneck formation, and shifts in traffic demand. As a result, the impact of construction activities is often non-linear and difficult to predict. Conventional traffic analysis methods, which typically rely on static models or localized assessments, are limited in their capacity to capture these dynamic interactions at scale. This limitation makes it difficult for planners and engineers to accurately evaluate the consequences of construction activities, compare alternative intervention strategies, or anticipate how disruptions will evolve over time.

In response to these challenges, there has been increasing interest in the use digital

twins that provide a structured framework for integrating elements into a unified analytical environment [6]. A digital twin of a transportation network can combine data, models, and simulation capabilities to represent the behaviour of the physical system under different conditions [7]. This allows for the exploration of hypothetical scenarios and supports the analysis of how changes to the network influence traffic flow and travel time.

This project presents a proof-of-concept implementation of a digital twin framework for the road network of Montreal, with a specific focus on analyzing the effects of construction-related disruptions on vehicle speed and travel time. The project operationalizes the framework through the development of a set of interconnected services that perform prediction, simulation, and visualization tasks. These services are designed to work together to provide a comprehensive view of network behaviour and to facilitate the analysis of construction impacts from multiple perspectives.

The primary contribution of this project lies in the design and implementation of an integrated digital shadow (digital twin prototype) that combines multiple analytical techniques within a single framework. Specifically, the project demonstrates how machine learning can be used to identify potential infrastructure issues, how graph-based models can be used to simulate network behaviour under changing conditions, and how visualization tools can be used to interpret and communicate results. The integration of these components highlights the potential of combining predictive and simulation-based approaches for analyzing transportation systems.

In addition to its technical contributions, the project establishes a modular architecture that can be extended in future work. While the current implementation focuses on historical data and simplified network modifications, the framework can be expanded to incorporate real-time data sources, more advanced traffic models, and additional system components which can be used to facilitate automated decision for improving road construction planning and implementation. This extensibility makes the proposed

approach a useful foundation for further research and development in the area of transportation system modelling.

2 Motivation

On August 10th 2025, the *Vérificatrice Générale* of Montreal released the results of 2024's yearly auditing [1]. One of the core sections addressed the management of road infrastructures, estimating a total of 4030 km in length of road worth 23 billion CA\$. Moreover, the audit evaluated an additional 3.3 billions required for repairs to restore paths to an *acceptable level*. In their efforts to sustain the road network, the city's administration has set various goals to assess the efficacy of their service - only to notice a lack thereof at the cost of production. Figure 2.1 shows the difference between the length of road where construction was planned (the blue bar) compared to the actual construction that was carried out and completed (orange bar).

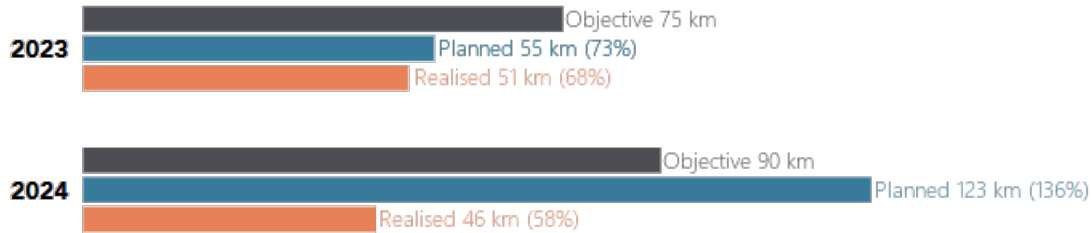


Figure 2.1: Insufficiency of maintenance work [1]

The cause of this issue was identified as the absence of a system or strategy meant to manage tasks between the administrations of the city, its municipalities and other agents. This would thus cause 1) difficulties in translating objectives to concrete actions, 2) create a lack of coordination and clarity in the distribution of responsibilities, 3) increase the risk of inefficacy in the maintenance of road infrastructures and 4) cause a loss of trust from the population [1]. But how could a digital twin address these issues? A fully automated system could precisely monitor the state of

the roads, order repairs by priority and inform responsible parties of the necessary steps for an efficient maintenance. In our estimate, it could significantly help with the difficulties 2), 3) and relatively help with the difficulties 1) and 4), due to a lack of procedure knowledge.

3 Data

When searching for data off which to base our services, we aimed for datasets which were reliable, comprehensive, current and made from a first-party source. We have aimed at the City of Montreal [8] and the Government of Quebec's [9,10] own database platforms to comply with the *first-party* requirement. Given the purpose of our DT, we searched for data related to road usage, road quality measurements, construction status, as well as general spatial information such as road length and connections.

We found a total of three datasets. The first is called **Obstacles and ongoing work** (rough translation), which includes data on construction site details such as duration, nature, location within the City of Montreal from 2021 to current time. This dataset meets the comprehensive and current requirements, reliability being affected by the irregular frequency of updates.

The remaining datasets, **Road segments for collecting travel times** and **Travel time on road segments**, were generated using Bluetooth sensors to detect devices at both the start and end of a road segment in sectors around Mount Royal and downtown Montreal. The second dataset holds the information regarding roads segments, each with their own unique string identifier, the concurrent sensors as well as the segment length. The third database holds the actual measurements: the times at which a device entered and exited the segment, the segment's ID as well as the date of the measurement, from which you can derive the speed and travel times. These latter datasets meet the comprehensive requirement, but holds data between 2016

and 2019 only, not to mention the periods within where sensors were disabled. This makes the second and third datasets less reliable and not so current.

Overall, the data available is less than we had hoped for. Furthermore, there is no clear mention of the method used to quantify the quality of the roads in the 2024 annual report. Our guess is the administration contracted a third-party to evaluate the repair costs within the road system, which leaves room for ambiguity. This means we'll have to, in one part, define what makes the *quality* of a road, or how we can calculate it. Secondly, we'll need to see if we can calculate said quality given the data available.

4 Modelling

How do we typically describe a road of low quality? One full of cracks and bumps. Though cracks and bumps are almost opposite by definition, they are both the result of the same cause: material which has been displaced by mechanical stress. We can define the displaced matter mathematically by:

$$\int_V |\rho_v(\vec{r}, t) - \rho_v(\vec{r}, 0)| dV$$

Where ρ_v is the volumetric density as a function of space-time and V is the volume of integration, in this case the road. If we were to use this formula for a simulation however, we would probably have to do a finite difference method over four thousand kilometres of road, which seems unachievable. Another way to simulate the deformation would be to define it as the sum of all past micro deformations - so, as we iterate through time, we measure the stress and evaluate the resulting deformation.

Classically, the sources of said mechanical stress on the road are attributed to the weight of the cars, the friction of the wheels and the ambient temperature. For those,

we make the following statements presented in Table 4.1.

Table 4.1: Factors of a Physics-based Simulation

Source	Orientation	Variables
Weight of car	Perpendicular to road	Mass of car & tire contact area
Wheel friction	Parallel to road	Speed & Acceleration
Temperature	Normal to road	Temperature gradient

The problem remains the same however: we don't want to have to simulate a large amount of road subsegment, but then we would lose on accuracy with each approximation. Do we neglect friction? Are all road segments the same temperature? Do all cars have the same weight and tire contact area? Do we have access to the necessary physics constants or data? And, as with many simulation, do we have an initial state?

Given the data available to us, it would seem the best we could is estimate an average amount of cars a road can be subjected to before needing repair. A physics-based simulation would require extra data, such as temperature, plastic deformation behaviours, distributions for car types on the road, which is simply out of our reach. That is why we will need to consider an alternative approach.

5 Digital Shadow Services

In this digital shadow, we present 3 services: 1) a prediction service that predicts whether a road segment requires construction or not, 2) a what-if service that allows a user to view the delay that would caused due to the modification of speed on a road segment, and 3) a dashboard visualization service using which the user can obtain the analysis and insight that we hope would benefit city planners and engineers. Each of these services are explained below in detail.

5.1 Prediction Service

The prediction service developed for the digital shadow aims to estimate whether a given road segment in Montreal will require construction within the next five years, based on historical speed and travel time data of the road segments. The service relies on data derived from publicly available datasets provided by the City of Montreal and the Government of Quebec. First, these datasets were ingested into a relational PostgreSQL database, where structured queries were used to compute descriptive and contextual features at the road segment level. Extracted features included average speed, average travel time, and their respective standard deviations, capturing both central tendencies and variability in traffic conditions. Additional spatial attributes were also calculated taking advantage of the Postgis feature that extends the capabilities of the PostgreSQL relational database by adding support for storing, indexing, and querying geospatial data. These attributes included geometric bearing of the road segment (expressed in degrees), the distance to the nearest construction activity, and the number of construction zones within a 100-meter radius. These features collectively form the input features for the prediction models.

A supervised dataset of 330 road segments was constructed, with each instance labelled according to whether that road segment currently has any on-going construction. On analysis, we identified that the dataset exhibited a class imbalance, with 235 non-construction instances and 95 construction instances. To mitigate this imbalance and prevent bias toward the majority class, the Synthetic Minority Over-sampling Technique (SMOTE) was applied during training. This method generates synthetic samples for the minority class (instances with construction), improving the model’s ability to learn meaningful decision boundaries.

Given the limited dataset size, three classical machine learning models were selected: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier.

Logistic Regression was included as a linear baseline model to assess whether a linear relationship exists between the engineered features and construction likelihood. The model was configured with a maximum of 1000 iterations to ensure convergence and incorporated class weighting set to “balanced” to account for class imbalance. The Random Forest Classifier was chosen to capture non-linear relationships through an ensemble of decision trees; it was configured with 100 trees (estimators), a maximum tree depth of 5 to control model complexity and reduce overfitting, balanced class weights, and a fixed random state for reproducibility. The Gradient Boosting Classifier was included as a more advanced ensemble method that iteratively improves prediction performance by focusing on previously misclassified samples; default hyperparameters were used for this model, allowing it to adaptively learn complex patterns without extensive tuning in this initial implementation.

The training procedure was designed to ensure robust evaluation despite the limited dataset size. The dataset was first partitioned into a training set and a holdout test set, with the holdout set reserved exclusively for final evaluation. The training data was evaluated using stratified 10-fold cross-validation to preserve class distribution across folds. Within each fold, SMOTE was applied only to the training subset to avoid data leakage, after which the model was trained and validated on unseen data. Prior to model training, feature scaling was performed using standardization, ensuring that all features had zero mean and unit variance, which is particularly important for models such as Logistic Regression.

After cross-validation, each model was retrained on the full training dataset (with SMOTE applied) and evaluated on the holdout set to obtain an unbiased estimate of performance. Table 5.1 quantifies the model performance using metrics such as precision, recall, F1-score, and accuracy, providing a comprehensive evaluation of classification effectiveness. In this setting, precision reflects the proportion of road segments predicted as requiring construction that truly correspond to construction

cases, while recall measures the ability of the model to correctly identify actual construction segments. The F1-score summarizes the trade-off between these two measures using their harmonic mean. Accuracy is also reported as an overall measure of correct classifications but can be less sensitive to minority-class performance. To further analyze model behaviour, feature importance scores (for Random Forest and Gradient Boosting) and coefficient analysis (for Logistic Regression) were used to interpret the influence of individual features.

Table 5.1: Model Performance (Confusion Matrix)

Model	Dataset	Construction	Precision	Recall	F1
Logistic Regression	CV	No	0.9528	0.9619	0.9573
		Yes	0.9701	0.9630	0.9665
		Acc.	0.9625		
	Test Set	No	0.9000	0.9000	0.9000
		Yes	0.9286	0.9286	0.9286
		Acc.	0.9167		
Random Forest	CV	No	1.0000	0.9905	0.9952
		Yes	0.9926	1.0000	0.9963
		Acc.	0.9958		
	Test Set	No	0.9091	1.0000	0.9524
		Yes	1.0000	0.9286	0.9630
		Acc.	0.9583		
Gradient Boosting	CV	No	1.0000	0.9905	0.9952
		Yes	0.9926	1.0000	0.9963
		Acc.	0.9832		
	Test Set	No	0.905	1.0000	0.9412
		Yes	1.0000	0.9132	0.9644
		Acc.	0.9553		

CV: Cross-Validation

Logistic Regression produced solid and consistent results, with an accuracy of 96.25% during cross-validation and 91.67% on the holdout set. Its precision, recall, and F1-scores are well balanced across both classes, suggesting that it provides stable and reliable predictions despite its relative simplicity.

Random Forest and Gradient Boosting demonstrated superior performance during cross-validation, both achieving an accuracy of 99.58% and 98.32% respectively. Their

class-wise precision, recall, and F1-scores are consistently close to 1.0, indicating excellent discrimination between the two classes. On the test set, both models maintained strong performance, achieving an accuracy of 95.83% and 95.53%, along with high precision and recall values for both classes.

Comparing the models, ensemble methods (Random Forest and Gradient Boosting) outperform Logistic Regression in terms of overall predictive metrics. However, all three approaches provide highly accurate and balanced predictions, making them suitable candidates for this classification task. The results suggest that incorporating more complex models can improve predictive performance, while simpler models like Logistic Regression still offer competitive and dependable results.

Moreover, to understand the drivers behind the models' predictions of whether a road segment requires construction, we extracted interpretability outputs (Figures 5.1, 5.2, 5.3). Across all three models, proximity to existing construction activity emerges as the dominant predictive factor. In both the random forest and gradient boosting models (Figures 5.2 and 5.3), variables related to construction proximity, particularly the distance to the nearest construction site and the presence of construction within 100 meters, account for the majority of feature importance. In comparison, traffic, speed, and geometric attributes contribute only marginally.

This pattern suggests that the models primarily capture a spatial dependency where road segments located near ongoing or recent construction are substantially more likely to require construction themselves. The logistic regression results (Figure 5.1) are consistent with this observation, with proximity-related variables exhibiting the largest-magnitude coefficients. These coefficients also provide directional information, indicating whether each feature increases or decreases the likelihood of construction under a linear additive formulation.

Unlike the tree-based models, logistic regression distributes weight more evenly across secondary features; however, their overall influence remains comparatively

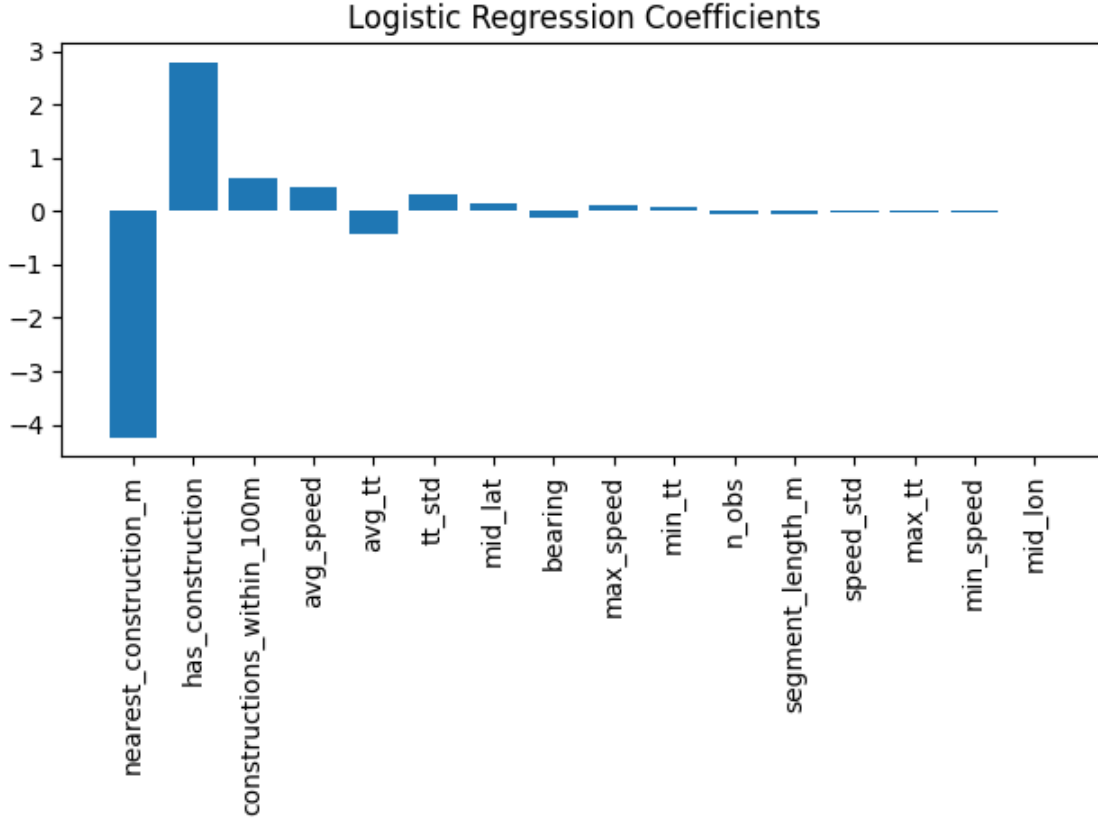


Figure 5.1: Magnitude and direction of feature coefficients in the logistic regression model.

limited.

5.2 What-If Service

The What-If Simulation Service is a scenario-based analytical tool built to help city planners assess the traffic impact of road construction before any physical work begins. At its core, the service represents the entire Montreal road network as a mathematical graph, where every road segment is a connection between two points, and the cost of travelling that connection is expressed as travel time. This structure allows the service to reason about how traffic flows across the city as an interconnected system rather than as a collection of isolated streets, meaning that a disruption on one segment can be traced through to its downstream effects on routes that may be several blocks away.

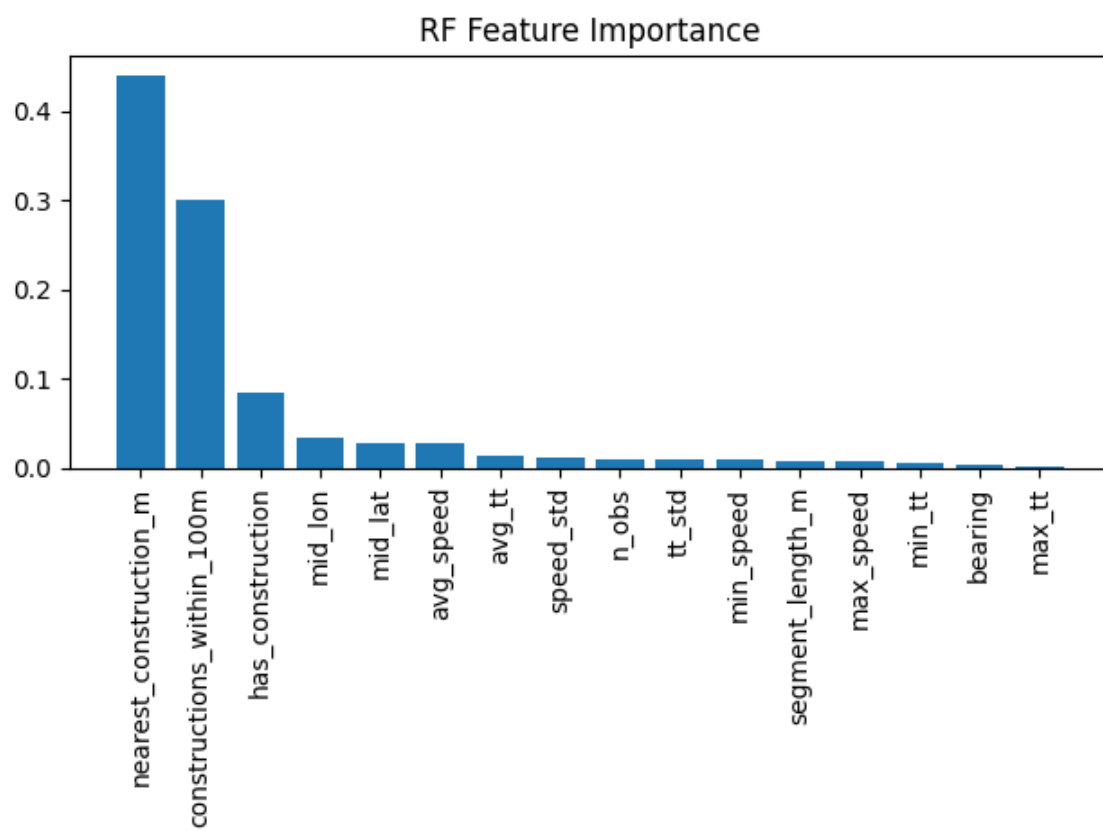


Figure 5.2: Relative importance of input features in the random forest model.

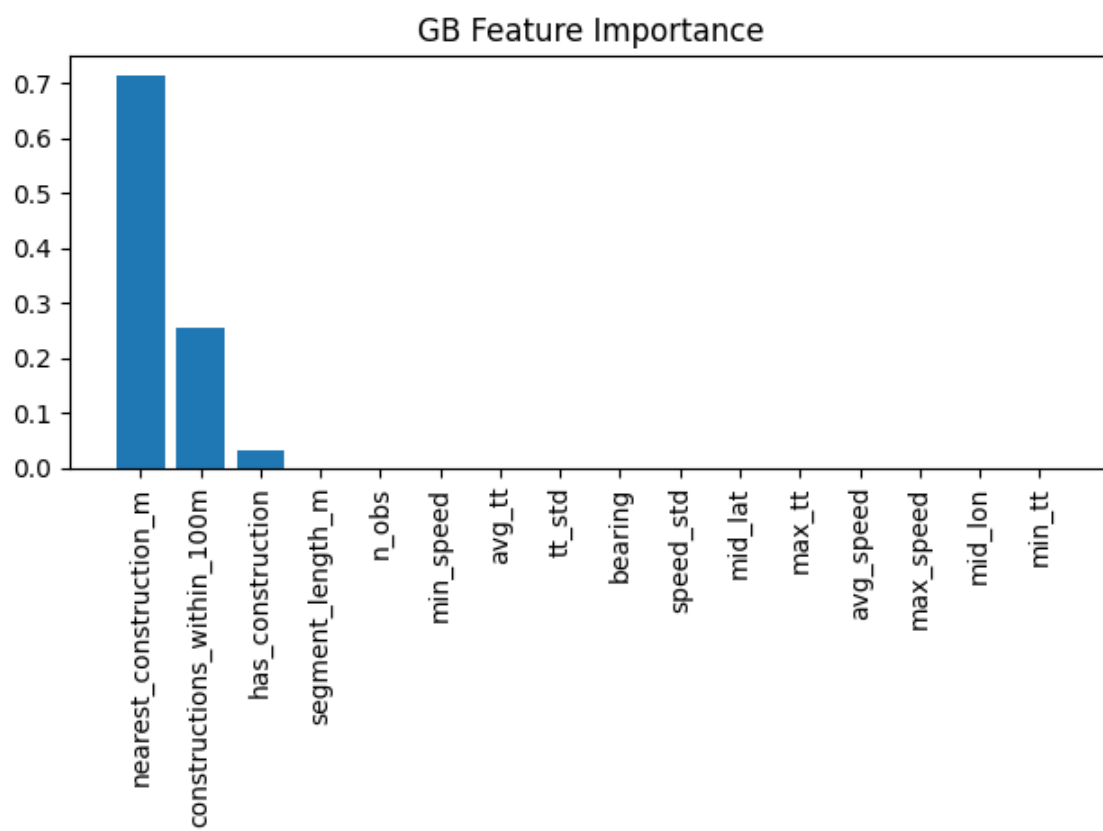


Figure 5.3: Relative importance of features used by the gradient boosting model.

The primary capability the service offers is the ability to simulate what happens to network-wide travel times when a specific road segment is slowed down or closed entirely. Consider a city planner is evaluating an upcoming construction project. This service allows them to select a segment and specifies the degree of disruption, which could be a partial speed reduction to reflect a construction zone with lowered speed limits, or a full closure to reflect a complete shutdown. The service then compares how traffic would route and how long journeys would take under normal conditions versus under the disrupted conditions. By examining a broad sample of origin-destination pairs across the network, it can estimate the average delay experienced across all trips that would have used or been rerouted around that segment, the worst-case delay for the most affected trips, the total cumulative delay across the network, and whether any origin-destination pairs become entirely unreachable. These metrics are summarized into an overall impact severity such as low, medium, or high, giving planners an immediate, interpretable signal of how consequential a proposed disruption would be.

The core analytical workflow is as follows. The service produces two versions of the road network graph, one with the baseline travel time representing current undisturbed conditions, and a modified graph in which the affected segment's travel time has been recalculated to reflect the reduced speed or, in the case of a full closure, the segment has been removed entirely. With these two graphs in hand, the service then applies a shortest-path algorithm (Dijkstra's algorithm) across a representative sample of origin-destination pairs drawn from across the network. For each pair, the algorithm independently finds the minimum-cost path, that is, the route with the lowest total travel time in both the baseline and modified graphs. The difference in path cost between the two graphs for a given pair represents the delay introduced by the disruption. If the optimal route in the modified graph costs more travel time than in the baseline, that excess is attributed as delay. In cases where a closure renders a previously reachable destination entirely inaccessible, the pair is recorded

as unreachable.

The practical advantage of this service is that it transforms what would otherwise be a largely intuitive and experience-driven planning judgment into a quantifiable, reproducible analysis. Rather than relying on a planner’s familiarity with a neighbourhood to guess whether closing a street will cause significant congestion, the service produces concrete delay estimates grounded in the actual geometry and connectivity of Montreal’s road network. This is particularly valuable when comparing multiple candidate construction timelines or routes, as planners can run successive scenarios and directly compare their projected impacts to identify the least disruptive option.

To evaluate the service in practice, three road segments were selected across different parts of the Montreal network and subjected to progressive speed reductions of 75%, 50%, and 25% of their baseline speed. For each scenario, the service computed the average network delay, the maximum delay observed across any single origin-destination pair, and the average per-trip delay in seconds. The results are summarized in Table 5.2.

Table 5.2: What-If Simulation Results Across Three Road Segments

Segment	Speed	Avg Delay	Max Delay	Avg Delay/Trip	Severity
Lorimier: Rosemont to Jean-Talon	75%	0.3%	15.0%	3.8s	Low
	50%	0.4%	19.9%	5.2s	Low
	25%	0.4%	19.9%	5.2s	Low
Rosemont: Saint-Michel to Papineau	75%	0.2%	13.9%	1.9s	Low
	50%	0.5%	30.4%	5.5s	Low
	25%	0.6%	38.1%	6.7s	Low
Notre-Dame: 6363 to Beauclerk	75%	0.3%	5.6%	3.8s	Low
	50%	0.9%	16.8%	11.5s	Low
	25%	2.8%	50.5%	34.6s	Low

Across all three segments, the overall severity remained low, suggesting that no single segment, when disrupted in isolation, produces a catastrophic network-wide

impact. However, the results reveal meaningful differences both across segments and across levels of disruption. For the Lorimier segment, average delays plateaued between the 50% and 25% speed scenarios, indicating that beyond a certain threshold of reduction, the network had already absorbed the disruption through rerouting and further speed degradation contributed little additional delay. The Rosemont segment exhibited a more gradual result, with average and maximum delays increasing consistently as the speed was reduced, and the Notre-Dame segment showed the most pronounced sensitivity where a 25% speed reduction produced a maximum delay of 50.5% and an average per-trip delay of 34.6 seconds, substantially higher than the other two segments under equivalent conditions.

This variation across segments underscores the utility of the service, where two segments subject to the same degree of construction-related disruption can produce very different network impacts depending on their position and role within the road network, and the service makes those differences visible and quantifiable ahead of time.

5.3 Dashboard Service

The dashboard visualization serves as the primary interaction interface between users and the digital twin of Montreal’s road network, enabling intuitive exploration, analysis, and simulation of road-related data and services. The dashboard is a web-based map visualization of the city of Montreal that displays the road segments and the current on-going road construction in the city. The web page includes sliders that allow the users to control change the time frame in order to observe traffic evolution over time on the road segments. In addition to displaying the historical traffic and current construction data, the dashboard also provides users access to outputs of the predictive services and what-if simulation. As such, it plays a key role in supporting infrastructure monitoring and decision-making.

The dashboard is designed to provide an intuitive and interactive environment for exploring the digital shadow. Its interface is structured around three main components: an interactive map, a control panel, and a legend section. Together, these elements allow users to visualize spatial information, interact with the proposed services, and interpret the displayed data efficiently.

The interactive map is the central component of the dashboard and provides a geographical representation of Montreal’s road network. Road traffic conditions are visualized using color-coded segments, while construction sites are represented as point markers with detailed information accessible on hover. The control panel centralizes all user interactions, including the sliders for temporal controls, toggles for viewing the historical traffic speed and construction, dropdown input for prediction model selection, and modifying the what-if simulation parameter. It also directly displays key metrics such as prediction accuracy of each machine learning model such as accuracy, precision, recall and F1-score, as well as, average delay, and maximum delay. Finally, the legend section provides a clear explanation of the visual encoding used on the map, improving the readability and interpretability of the displayed data.

The dashboard is implemented as a native web-based application following a client–server architecture. The frontend is developed using Angular [11], which is a Typescript-based web framework that allows modular and effective implementation of a frontend, and managing the different elements of the interface. The frontend is further enhanced by integrating a map rendering and spatial data visualization that are handled using the Leaflet.js library [12], which supports efficient manipulation of multiple geospatial layers.

User interactions within the dashboard trigger event-driven updates that propagate through the frontend components and initiate HTTP requests to the backend API services. The backend API server is implemented in Flask [13], a python-based library, and acts as the central layer responsible for data processing, model inference, and

simulation execution. It exposes dedicated endpoints for accessing processed traffic data, construction datasets, prediction results, and what-if analysis outputs. Data persistence is ensured through a PostgreSQL database, which stores both historical traffic information and construction records. The backend performs SQL-based queries to retrieve and preprocess the required subsets of data before returning them to the frontend in a structured format.

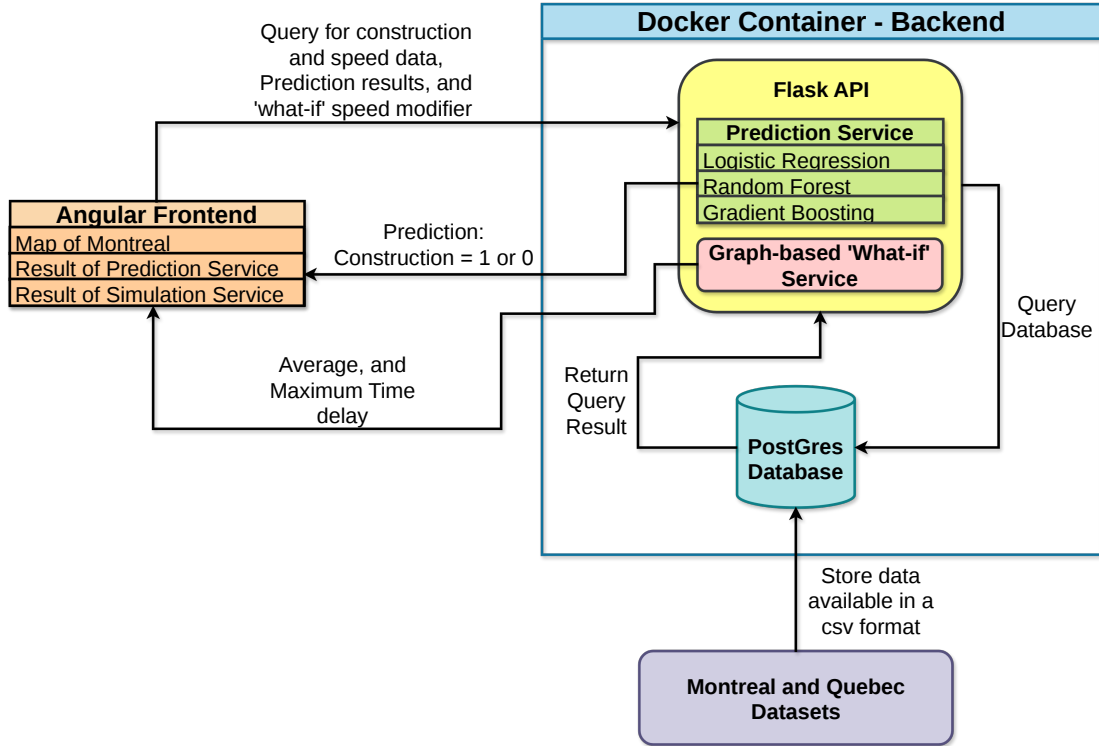


Figure 5.4: Connection and flow of data between the Angular-based frontend, Flask-based backend API, and PostGRES database that are containerized in a docker container

Figure 5.4 illustrates the interaction between the various system components and how they collectively deliver analysis and insights to the user through the web-based frontend. Road segment, speed, travel time, and construction data, provided in CSV format, are first ingested and stored in the PostgreSQL database. The Flask server then retrieves data from the database and trains three machine learning models to predict whether a road segment will require construction. Once trained,

the predictions generated by these models are exposed through the Flask server via API endpoints. In addition, the Flask server is responsible for constructing a graph from database queries, which is subsequently used by the what-if analysis service. The is also exposed to the frontend for querying using an API endpoint. The backend components described above are containerized using Docker to package all dependencies required for deployment and execution of the backend system. The Angular frontend consumes the available API endpoints to retrieve data, which is then visualized on a Leaflet-based map. It displays the machine learning models predictions, ground truth labels from the prediction service, and results from the what-if analysis, including estimated delays caused by simulated speed adjustments.

6 Digital Shadow Architecture & Modelling

The developed digital shadow adopts a modular architecture that integrates data acquisition, modelling, simulation, and visualization within a unified framework. The system follows a layered structure typical of digital twin architectures, separating the physical domain, enabling technologies, data and models, and service execution.

The overall architecture of the system is illustrated in the digital twin constellation shown in Figure 6.1.

As shown in Figure 6.1, the architecture is organized into distinct but interconnected layers. On the left, the physical domain consists of the Montreal road network, active road construction sites, and the system operator, represented by the city planning engineer. These elements generate the raw information required for the digital representation, including speed data, construction activity, road network geometry, and travel time.

This information enters the digital environment through the enabling layer, which includes communication and interface technologies such as WiFi, HTTPS, and Bluetooth.

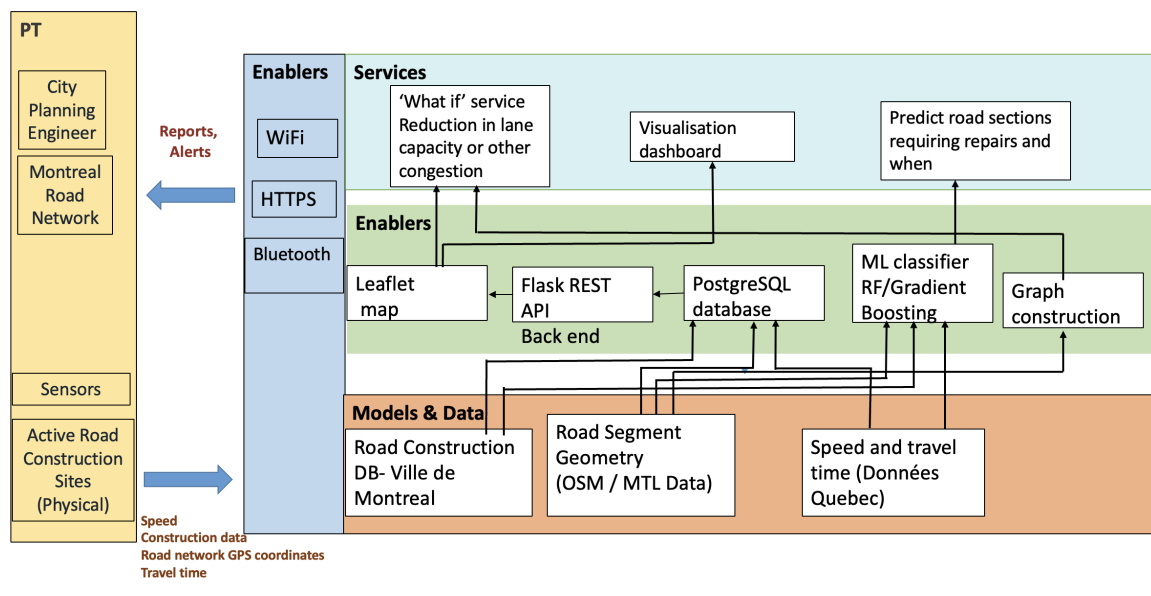


Figure 6.1: Architecture of the developed MonDTreal digital shadow, illustrating the interaction between the physical system, enabling technologies, data and models, and service layer.

While these components represent how data would be transmitted in a fully deployed system, the current implementation relies primarily on historical datasets.

The lower section of the architecture corresponds to the models and data layer. This includes the road construction database, road segment geometry derived from OSM and Montreal datasets, and speed and travel time data obtained from Données Québec. These datasets form the foundation of the digital shadow, providing the inputs required for both predictive modelling and simulation.

Above this layer, the backend and computational components process and organize the data. A PostgreSQL database with PostGIS extensions manages storage and spatial queries, while a Flask REST API provides a structured interface for communication between components. Additional modules, such as the graph construction component and the machine learning models (Logistic Regression, Random Forest, and Gradient Boosting), operate on this data to generate predictions and support simulation.

At the top of the architecture, the service layer provides the main functionalities of the system. These include the prediction service, which identifies road segments likely

to require construction, the what-if simulation service, which evaluates the impact of disruptions such as reduced lane capacity, and the visualization dashboard, which presents results to the user in an interactive format.

Finally, the outputs of these services are communicated to the operator in the form of reports, alerts, and visual insights. While the system provides decision-support capabilities, it does not directly influence the physical system. There is no automated feedback or control mechanism linking the digital system back to the physical domain.

This layered organization highlights the modular and extensible design of the system. Each component performs a distinct role while remaining connected through well-defined interfaces. Figure 6.2 shows the transfer of data between the physical twin and its digital counterpart. The noticeable absence of a concrete automated bidirectional communication from the digital twin to the physical twin confirms that the implemented system operates as a digital shadow rather than a fully realized digital twin.

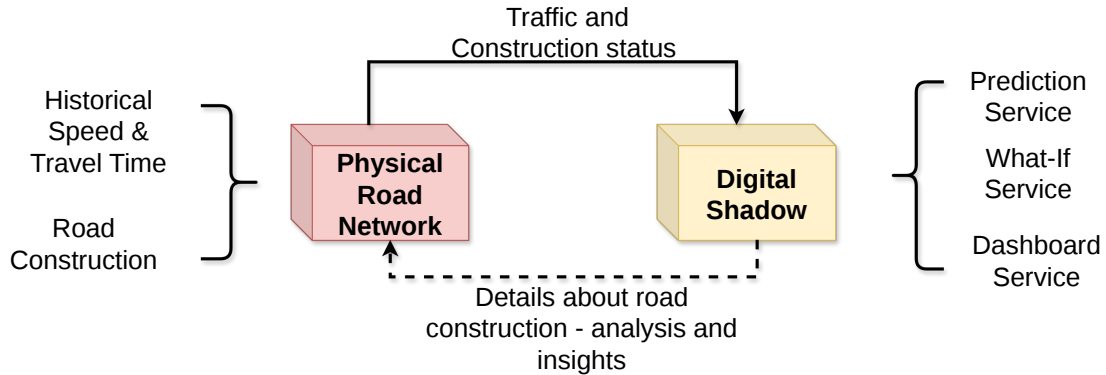


Figure 6.2: Transfer of data between the Physical Twin and the Digital Shadow

7 Digital Shadow Insights & Characteristics

The developed digital shadow has numerous characteristics consistent with digital twin frameworks particularly the ISO 23247. A key characteristic is the integration of heterogeneous data sources. Traffic data, construction records and spatial road information are combined to form a coherent representation of the system, enabling analysis of relationships between infrastructure conditions and traffic performance.

The system also combines predictive and simulation based modelling. Machine learning models provide predictions of future construction needs, while the graph based model supports scenario analysis through what-if simulations. This dual capability reflects the analytical and predictive roles of the digital shadow.

Modularity is another important feature. The architecture separates data management, modelling, and service layers, allowing components to operate independently while remaining connected through a central API. This improves scalability and maintainability.

These characteristics are illustrated in the digital twin constellation shown in Figure 7.1. The constellation highlights the flow of information from the physical asset

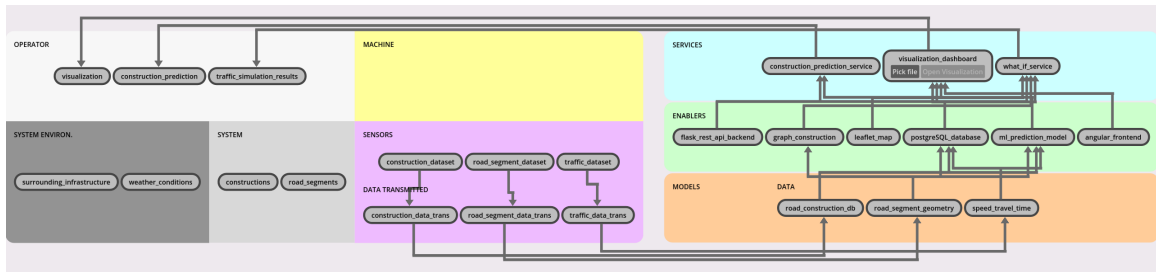


Figure 7.1: Digital shadow constellation showing system architecture and data flow.

to the digital system. Physical elements such as road infrastructure and construction activities are represented through datasets, processed via enabling technologies, and used by higher level services to generate insights.

As mentioned earlier, a key observation is the absence of an automated feedback loop from the digital shadow to the physical twin . Data flows from the physical system to the digital system, but no automated control actions are returned. Under ISO 23247, this unidirectional flow classifies the system as a digital shadow rather than a full digital twin.

To systematically position MonDTreal within digital twin engineering, Table ?? summarizes the system against a complete set of digital twin characteristics. A comprehensive and interactable reporting page that describes the digital shadow can also be found [here](#).

Table 7.1: DT Characteristics

	Characteristics	Description
C1	System under study	Digital twin system for Montreal road network monitoring and decision support
C2	Physical acting components	constructions: Road areas affected by ongoing construction or maintenance work road_segments: Road sections used for vehicle circulation
C3	Physical sensing components	traffic_dataset construction_dataset road_segment_dataset
C4	Physical-to-virtual interaction	traffic_data_trans road_segment_data_trans construction_data_trans
C5	Virtual-to-physical interaction	The DT provides insights and scenario analysis that support decision-making for road planning and construction management.

	Characteristics	Description
C6	DT services	visualization_dashboard: Web-based visualization service providing interactive map and dashboard for monitoring road network state and simulations construction_prediction_service: Predictive analytics service using historical and spatial data to estimate future road construction needs and impacts what_if_service: Interactive simulation service that computes traffic scenarios under different road closure or disruption conditions
C7	Twinning time-scale	The DT is synchronized with updated road construction and traffic datasets. Interactive computation with near-instant response time based on static or preloaded data.
C8	Multiplicities	The current implementation represents a single Digital Twin instance of the Montreal road network, although the framework could be extended to support multiple instances for different cities or scenarios.
C9	Life-cycle stages	The DT supports the design and the service phases. In the service phase, it supports creating, executing, saving, analyzing and terminating.

	Characteristics	Description
C10	DT models and data	Models: logistic_regression: Linear baseline model predicting road segment construction likelihood using engineered traffic and spatial features random_forest: Ensemble of 100 decision trees (max depth 5) for non-linear construction prediction with balanced class weighting gradient_boosting: Iterative ensemble model for complex pattern learning in construction prediction graph_dijkstra: Graph-based shortest-path model used to simulate travel time under road disruption scenarios Data: road_construction_db: Structured dataset containing road construction activities, including work zones and schedules road_segment_geometry: Geospatial dataset describing road network topology speed_travel_time: Traffic performance dataset containing speed and travel time measurements across road segments over time

	Characteristics	Description
C11	Tooling and enablers	angular_frontend: Web frontend framework responsible for rendering the dashboard and user interface flask_rest_api_backend: REST API backend providing data access, processing, and orchestration for services graph_construction: Graph-based processing engine used to model and analyze road network relationships for scenario simulation leaflet_map: Interactive map library used to display road network layers, traffic segments, and construction zones ml_prediction_model: Machine learning models used to predict construction needs and traffic impact patterns postgresql_database: Relational database (with PostGIS extension) storing structured road network data, traffic information, and simulation results
C12	DT constellation	The system integrates multiple software components including an Angular frontend, Leaflet-based map visualization, Flask REST API backend services, PostgreSQL/PostGIS data storage, and machine learning prediction modules.

	Characteristics	Description
C13	Twinning process and DT evolution	The Digital Twin was developed incrementally: collection of open datasets related to road networks, traffic, and construction; integration and structuring of geospatial data; development of data storage and processing pipelines; implementation of visualization and what-if simulation services; integration of machine learning models for construction prediction; and continuous refinement of data quality, simulation accuracy, and user interaction capabilities.
C14	Fidelity and validity considerations	The fidelity relies on publicly available datasets including road geometry, traffic conditions, and construction information. The road network is represented with accurate geospatial data, while traffic and construction data are periodically updated but may not reflect real-time conditions. Simulation and prediction services provide approximate results based on historical data and learned patterns, intended to support decision-making rather than provide exact forecasts.
C15	DT technical connection	The frontend communicates with the backend over HTTP REST APIs, while the backend accesses the PostgreSQL database and prediction services.
C16	DT hosting/deployment	The DT is deployed as a web-based application accessible through a browser, with backend components containerized using Docker.

	Characteristics	Description
C17	Insights and decision making	construction_prediction: Prediction of future construction needed based on ML model outputs traffic_simulation_results: What-if simulation of traffic under different road closure or speed reduction scenarios visualization: Visualization of road traffic and construction events on an interactive map
C18	Horizontal integration	The Digital Twin integrates multiple software components across the same operational layer, including a web-based frontend, backend services, data storage, and machine learning modules. It also connects to external open datasets related to road networks, traffic conditions, and construction activities, enabling consistent data exchange and coordinated functionality.
C19	Data ownership and privacy	Datasets have been provided online to the public via Données Québec and City of Montreal open data platforms. No privacy-related data are stored.
C20	Standardization	Communication between services is performed through RESTful HTTP APIs and geospatial data formats compatible with PostGIS and Leaflet.

	Characteristics	Description
C21	Security and safety considerations	The Digital Twin ensures security through standard web protocols, including HTTPS for secure communication between frontend and backend services. No sensitive personal data is stored as the system relies on publicly available datasets. Outputs are intended for decision support rather than real-time automated actions, reducing risks associated with incorrect predictions or simulations.

Despite these limitations, the system demonstrates the value of digital twins as decision support tools, particularly for exploratory analysis and infrastructure planning. Finally, the system demonstrates the potential of digital twins as decision support tools. With improved data integration and real-time capabilities, the system could evolve into a fully functional digital twin.

8 Challenges in Digital Twin Engineering

8.1 Data

One of the primary challenges encountered in this project was related to the availability, consistency, and quality of data required to develop the digital twin services. The effectiveness of the proposed digital shadow relies heavily on the integration of multiple datasets, including historical traffic data (speed and travel time) and road construction data. However, several limitations in these datasets introduced significant constraints on the development and evaluation of the system.

A major issue arose from the lack of temporal alignment between datasets. The

historical speed and travel time data used in this project was collected between 2016 and 2019, whereas the road construction dataset was only available for a much later period, spanning from 2021 to 2026. This temporal mismatch created a fundamental limitation, as it was not possible to directly correlate traffic conditions with construction activities. As a result, it remained unclear whether the observed speed and travel time patterns in the historical dataset were influenced by construction-related disruptions or reflected normal traffic behaviour. This ambiguity limited the ability to accurately model or quantify the real-time impact of construction on traffic conditions in order to create an accurate digital twin.

This issue also affected the training and validation of machine learning models. Since the ground truth regarding construction impact was not available for the same time period as the traffic data, it was difficult to confidently label or interpret patterns within the dataset. Consequently, predictions generated by the model could not have been directly validated against real construction events, reducing the reliability of the predictive service.

An additional challenge related to the traffic dataset was the presence of missing and incomplete data. For example, in certain years such as 2018, speed and travel time data were not recorded for specific months, including January, February, and December. No explanation was provided for these omissions. These gaps reduced the temporal continuity of the dataset and required careful handling during preprocessing and analysis. The absence of data for extended periods also limited the ability to capture seasonal variations in traffic patterns, which may be relevant when analyzing transportation systems.

We spent approximately 5 weeks identifying issues and looking for alternatives. However, after a thorough search, we concluded that these dataset are the only ones good enough for our project, as we couldn't find any other datasets that fitted our needs for the project. Moreover, in attempts to address these limitations by

sourcing alternative data introduced further complications. External providers, such as commercial traffic data platforms (e.g., TomTom), were considered as potential sources of more recent or continuous data. However, accessing such data at the required temporal resolution would have required frequent API requests or automated scraping, potentially violating usage policies and rate limits. This raised both technical and ethical concerns, making it impractical to rely on such sources for this project. As a result, the study was constrained to publicly available datasets despite their limitations.

The road construction dataset presented its own set of challenges. First, the dataset contained heterogeneous construction records, including not only roadwork but also other types of infrastructure or urban development activities. The absence of explicit labels distinguishing road construction from other construction types required manual inspection and filtering. This process involved analyzing textual descriptions and metadata to identify relevant entries, which was time-consuming and prone to ambiguity. Ensuring that only road-related construction events were retained was necessary for maintaining the integrity of the analysis.

Furthermore, the construction dataset exhibited non-periodic update intervals, which complicated its integration into a system intended to approximate real-time behaviour. Updates to the dataset did not follow a consistent schedule; in some cases, new data was added multiple times within a single day, while in other instances, updates were delayed by several days. This irregularity introduced challenges in data ingestion and synchronization, as it was difficult to determine when the dataset reflected the most current state of construction activities.

These challenges related to data availability, temporal misalignment, incompleteness, heterogeneity, and irregular updates had a significant impact on the design and implementation of the digital twin. They necessitated careful preprocessing, constrained certain aspects of the digital twin development, and influenced the overall scope of

the project limiting our implementation to a digital shadow.

8.2 Simulation vs Neural Network vs Machine Learning Model

A key challenge in this project was the selection of an appropriate modelling approach for predicting whether a road segment requires construction or maintenance. Three broad approaches were considered: physics-based simulation, neural networks, and classical machine learning models. Each approach presents distinct advantages, but also important limitations within the context of this project.

Physics-based simulation models were initially considered due to their ability to explicitly represent the mechanisms underlying road degradation. Such models typically incorporate detailed parameters including material properties (e.g., asphalt quality), vehicle characteristics (e.g., weight and load distribution), environmental conditions (e.g., temperature and precipitation), and interaction effects such as tire–road friction. However, the implementation of such models requires access to comprehensive and high-resolution data describing these factors. In practice, acquiring this data is challenging due to limited availability, proprietary restrictions, and potential ethical concerns surrounding data sharing. As a result, a physics-based approach was deemed infeasible for this project.

Given these constraints, data-driven approaches were explored, specifically neural networks and classical machine learning models. Neural networks are known for their strong predictive capabilities, particularly in complex and non-linear problem settings. However, their effectiveness is highly dependent on the availability of large and diverse datasets. In this project, although historical speed and travel time data were available at high temporal resolution, the spatial coverage was limited to approximately 330 road segments. This relatively small feature space increases the risk of overfitting when training neural networks, leading to poor generalization on unseen data. Additionally, neural networks typically function as black-box models,

providing limited interpretability. This lack of transparency poses a challenge in understanding the underlying factors contributing to predictions, which is an important requirement in infrastructure-related decision-making contexts.

In contrast, classical machine learning models offer a more suitable balance between performance and interpretability under data-constrained conditions. These models are generally less prone to overfitting with smaller datasets and allow for the extraction of feature importance and other explanatory insights. This interpretability is critical for identifying the key factors influencing road degradation, thereby supporting more informed analysis and decision-making.

Based on these considerations, classical machine learning models were selected for the prediction service. This choice reflects a pragmatic trade-off between data availability, model complexity, predictive performance, and interpretability. However, despite machine learning achieving relatively high predictive accuracies of 91% and 95% for identifying road segments requiring construction, an important challenge still remains in managing the rate and impact of false positives. In this context, a false positive corresponds to incorrectly classifying a road segment as requiring construction when it does not. Given the substantial financial costs and resource allocation involved in road construction projects, such errors carry significant consequences, potentially leading to misdirected investments and inefficient use of limited budgets. Therefore, overall accuracy alone is not a sufficient performance metric, and further refinement of the models is necessary to reduce false positives and improve decision reliability.

8.3 What-If Simulation

Another significant challenge encountered in this project relates to the design and accuracy of the what-if simulation service. While the service provides a functional framework for analyzing the impact of construction-related disruptions, it relies on

simplifying assumptions that limit its ability to fully capture real-world traffic dynamics.

The simulation models the road network as a graph, where nodes represent segment endpoints and edges correspond to road segments weighted by travel time. Travel time is computed as a function of segment length and average speed. Within this framework, users can modify the speed of selected road segments to simulate disruptions such as construction. The impact of these changes is then evaluated by comparing shortest paths, computed using Dijkstra’s algorithm, between the baseline and modified graphs, allowing for the estimation of delay.

However, this approach assumes that changes in speed affect only the directly modified road segment. In reality, traffic systems exhibit strong interdependencies, and localized disruptions often produce cascading effects across the network. For example, a reduction in speed or partial blockage on one segment can lead to congestion, rerouting behaviour, and increased load on adjacent or alternative routes. These secondary effects are not captured in the current implementation, resulting in an oversimplified representation of network behaviour.

Additionally, the relationship between construction activities and speed reduction is not deterministic. Factors such as lane closures, traffic control measures, driver behaviour, and time-of-day variations all influence how traffic flow is affected. The current model addresses this uncertainty by allowing users to manually adjust speed via a slider; however, this introduces a level of subjectivity and does not guarantee realistic or consistent outcomes. Consequently, the resulting delay estimates that are expressed as average delay, maximum delay, and total time delay should be interpreted as approximations rather than precise predictions.

Furthermore, the service also omits several important traffic-related factors, including congestion levels, traffic density, signal timing, and dynamic demand patterns. The use of static shortest-path computation does not account for adaptive routing behaviour or temporal evolution of traffic conditions. These limitations highlight the gap between

the simplified simulation and real-world traffic systems.

Addressing these challenges would require access to richer datasets, including real-time traffic information and detailed road usage patterns, as well as the integration of more advanced traffic modelling techniques. Incorporating domain expertise in transportation engineering would also be essential to better represent phenomena such as congestion propagation and network-wide flow redistribution.

Despite these limitations, the what-if simulation service serves as a proof of concept, demonstrating how graph-based models can be used to explore the potential impacts of localized disruptions. While not fully representative of real-world conditions, the tool provides a useful starting point for scenario analysis and illustrates how such systems could support urban planning and decision-making with further refinement.

8.4 Engineering Process

A major challenge in this project was the practical realization of a fully operational digital twin, particularly with respect to establishing a closed-loop interaction between the digital and physical systems. While the conceptual framework of a digital twin assumes automated bidirectional communication, where the digital system both receives data from and actively influences the physical system, this level of integration proved difficult to achieve in practice.

Although the project successfully incorporates data from the physical road network into the digital environment, completing the feedback loop in the opposite direction remains a significant limitation. Specifically, the system lacks a direct mechanism to translate analytical outputs into physical actions. In the context of road construction and maintenance, such actions inherently require human intervention. For example, even if the system identifies segments requiring maintenance or proposes optimized interventions, these recommendations must be evaluated, approved, and executed by human stakeholders. Unlike domains where automated control systems can directly

actuate physical processes, urban infrastructure management depends on manual decision-making and operational constraints. As a result, the system functions as a digital shadow rather than a true digital twin, with an indirect and non-automated connection to the physical system.

In addition to this limitation, the development of a fully integrated system is complicated by the involvement of multiple stakeholders across different domains. Urban transportation systems are managed by various governmental departments, contractors, and regulatory bodies, each with their own priorities, data standards, and operational procedures. Coordinating across these entities to establish a unified engineering workflow is inherently complex and requires substantial collaboration. Ensuring that a system satisfies the requirements of all stakeholders while maintaining consistency and interoperability presents a non-trivial challenge. Furthermore, this issues propagated and is relevant to the group dynamics as this project required an individual whose expertise lied in the field of civil engineering. However, due to the lack of any such individual, we had to rely on our own knowledge for this project.

Another important constraint is the availability and timeliness of data. A fully functional digital twin relies on continuous, real-time data streams to accurately reflect current system conditions and support responsive decision-making. However, access to such data is often limited by infrastructure, policy, and technological barriers. Implementing real-time data pipelines would likely require significant investment and institutional support, including the deployment of new data collection systems and agreements with public agencies. In the absence of these capabilities, the system must rely on historical or delayed data, reducing its ability to operate as a real-time decision-support tool.

These challenges highlight that the creation of a true digital twin extends beyond technical implementation and into organizational, infrastructural, and operational domains. Consequently, this project adopts a proof-of-concept approach, demonstrating

the feasibility of integrating predictive modelling, simulation, and visualization within a unified framework, while acknowledging that substantial additional development and collaboration would be required to achieve a fully realized digital twin system.

9 Conclusion

This project presents a digital shadow for analyzing the impact of road construction on traffic conditions in Montreal. The main contribution of this work lies in the design and implementation of an integrated framework that combines machine learning, graph-based modelling, and visualization within a single system. Rather than addressing these components independently, the project demonstrates how predictive analytics and network simulation can be coupled to provide a more comprehensive understanding of transportation system behaviour.

First, data (road segment, speed, travel time, and construction) are ingested from various open data sources and stored in a PostgreSQL database. Utilizing this data, three digital shadow services are presented, 1) A prediction service, 2) a what-if service, and 3) a dashboard visualization service.

The prediction service enables the identification of road segments that are likely to require construction based on historical traffic patterns. This provides a data-driven approach utilizing 3 machine learning model to assess infrastructure, allowing potential problem areas to be highlighted without relying solely on manual inspection. The what-if simulation service extends this capability by allowing users to evaluate the impact of localized disruptions, such as reduced speeds or partial closures, on network-wide travel time using a graph-based modelling approach. By comparing baseline and modified scenarios, the system provides quantitative insight into how disruptions propagate across the network. The visualization dashboard integrates these functionalities into an accessible webpage-based interface, enabling users to

explore results interactively and interpret complex outputs more effectively.

From a practical perspective, these capabilities offer clear benefits for city planners and transportation stakeholders. The system can support proactive maintenance planning by identifying segments that may require intervention, assist in evaluating alternative construction strategies before implementation, and provide a structured way to communicate the expected impact of disruptions. In this sense, the digital shadow serves as a decision-support tool that complements existing planning processes, particularly in environments where data is available but not fully utilized.

Approximately 150 hours were spent on developing this proof-of-concept project within a limited timeframe, with emphasis placed on demonstrating feasibility and integration rather than achieving full operational deployment. Within this scope, the system successfully captures key elements that are required to create a fully functioning digital twin, including data integration, modelling, and service-based interaction, while remaining implementable using publicly available datasets and standard tools.

The proposed digital shadow system has several limitations preventing it from being a digital twin. The most notable limitation is the temporal mismatch between historical traffic and construction datasets, which prevents direct correlation of traffic patterns with construction activity. The lack of real-time data also limits the model to offline analysis, while the graph-based simulation depends on simplifying assumptions that exclude congestion spread, traffic density, and adaptive routing. Despite this, the project provides a strong foundation for applying digital twin concepts to urban transportation, showing that data-driven prediction and network-based simulation

can still produce useful insights and guide future improvements.

10 Future Works

Building on the limitations identified in the current implementation, future work will focus on extending the system toward a fully functional digital twin, both in terms of technical capabilities and operational scope.

A primary direction is the integration of real time data sources to enable continuous synchronization between the physical and digital systems. This would involve incorporating live traffic feeds, sensor data and dynamically updated road construction information. Such an extension would address the current reliance on historical datasets and significantly improve the system’s ability to reflect real world and current conditions. This could potentially be done by collaborating with the respective government organizations and restarting the data collection programs using which the historical data was obtained.

Furthermore, the simulation framework will also be enhanced to overcome the limitations of the current graph based approach. Future developments will incorporate more realistic traffic dynamics, including congestion propagation, flow interactions, and adaptive routing behaviour, allowing the system to better capture network wide effects of localized disruptions.

In addition to this, physics-based modelling will be incorporated where applicable. This includes modelling road degradation, traffic–infrastructure interaction, and environmental effects such as temperature and precipitation. Combining these physics-informed models with data-driven approaches would improve both interpretability and predictive reliability.

The prediction component will be further improved by expanding the dataset to include additional features such as weather conditions, traffic volume, road material

properties, and maintenance history. Addressing the temporal misalignment between datasets will also be critical for improving model validity. With richer datasets, more advanced modelling approaches, including neural networks and hybrid models will be explored.

Moreover, the proposed digital shadow can be extended beyond analysis to support planning and operations by linking its outputs to decision-making workflows. It could automatically identify candidate road segments for intervention, assist in generating Requests For Quotations (RFQs) and Bills of Quantities (BoQs), and integrate cost models and historical data to improve budget planning, prioritization, and resource allocation.

Further development includes scheduling and optimization capabilities, where prediction and simulation outputs are used to create construction plans that balance cost, duration, and traffic impact. Introducing a feedback mechanism would allow the system to generate prioritized recommendations and alerts, moving closer to a functional digital twin.

Finally, the framework can be standardized and scaled to other cities by generalizing data pipelines, models, and architecture. Together, these extensions transform the current system into a scalable, data-driven decision support platform for end-to-end urban infrastructure management.

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A Source Code

The source code for the project can be found on GitHub: github.com/HamzaGiTX786/LOG6310-Course-Project-Road-System-DT

B Video

A video showing the digital shadow, and explaining its services, giving a viewer clear sense of what the system does and what value it provides can be found on YouTube: [YouTube video](#)